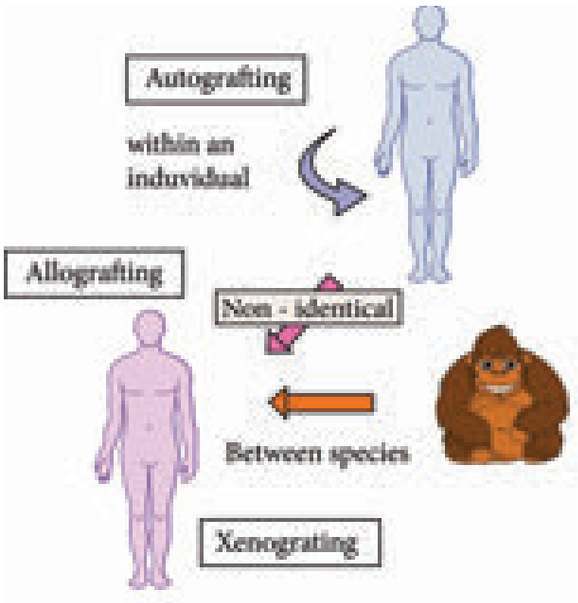


Figure 11.11: Types of Grafting

Graft Acceptance

When transplantation is made between genetically identical individuals the graft survives and lives as healthy as it is in the original places. When the graft tissue remains alive, it is said to be accepted and the process is called graft acceptance.

Graft Rejection

When transplantation is made between genetically distinct individual the graft tissue dies and decays. When the graft tissue dies, the graft is said to be rejected and the process is called graft rejection. It is of two types. They are

i. Host Verses Graft Reaction

ii. Graft Verses Host Rejection.

Host Verses Graft Reaction (HVG)

The graft tissue antigens induce an immune response in the host. This type of immune response is called host versus graft reaction.

Allograft Rejection

Types of allograft rejection

- Acute rejection–Quick graft rejection. It is due to stimulation of thymocytes and B lymphocytes
- Hyperacute rejection–It is a very quick rejection. It is due to pre-existing humoral antibodies in the serum of the host as a result of presensitization with previous grafts.
- Insidious rejection–It is a secret rejection due to deposition of immune complex on the tissues like glomerulus membrane that can be demonstrated in kidney by immune fluorescence.

Mechanism of Allograft Rejection

Immunological contact

When tissue is implanted the graft as can pass into local lymph nodes of the host. The graft antigens then make contact with the lymphocytes of the host. Production of sensitized T cells and cytotoxic antibodies are produced in the host. This brings about graft rejection.

First set rejection

When the graft is made between genetically different individuals, the graft gets blood supply from the host and it appears to be normal for the first 3 days. But on the 5th day, sensitized T cells, macrophages and a few plasma cells invade the graft. Inflammation starts in the graft. This leads to necrosis. It is similar to the primary immune response to an antigen.

Second set rejection

When a graft is implanted in an individual who has already rejected a graft is second set rejection. This is similar to the secondary immune response of our body.

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